MARKET REACTION FINGERPRINT

Market Reaction Fingerprint

Investigating Stock-Specific Responses to Financial News Sentiment

Mehak Arora
Indira Gandhi Delhi Technical University for Women (IGDTUW)
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Module A alignment: A tactical, high frequency stock-index rebalance

THE QUESTION

Does Financial News Sentiment Relate to Future Stock Returns?

Problem

- Financial news can influence investor sentiment and market expectations.
- Individual stocks may respond differently to similar sentiment signals.
- A sentiment signal does not automatically translate into a reliable or profitable trading strategy.

Research question

Can stock-specific relationships between financial news sentiment and future returns be identified, validated, and evaluated in a portfolio backtest?



Illustrative only — not measured results.

Proposed approach

1. Analyze financial news with FinBERT.
2. Aggregate sentiment by stock and trading date.
3. Measure correlations with future stock returns.
4. Evaluate a sentiment-based portfolio against an equal-weight benchmark.

From Unstructured Financial News to Portfolio Analytics

- Financial news: FNSPID NASDAQ dataset from Hugging Face, containing article text and stock symbols.
- Historical market data: closing prices for 11 stocks, covering January 2018 to January 2024.
- Data preparation: data loading, cleaning, and alignment to available trading dates.
- Sentiment engine: FinBERT (ProsusAI/finbert), producing positive, negative, and neutral sentiment probabilities.
- Daily aggregation: aggregate sentiment signals by stock and trading date.
- Reaction fingerprint: calculate correlations between sentiment signals and future single-day returns at 1, 2, 3, and 5 trading-session lags; apply chronological train-test validation and false discovery rate correction.
- Portfolio evaluation: construct portfolio weights and perform transaction-cost-aware backtesting against an equal-weight benchmark.
- Application: interactive Streamlit analytics dashboard.

METHOD

Measuring Stock-Specific Sentiment Relationships

A. Sentiment scoring

Use FinBERT to obtain positive, negative, and neutral probabilities.

$$Sentimentscore = P(positive) - P(negative)$$

B. Reaction fingerprint

- Evaluate daily sentiment score and change in sentiment.
- Compare each signal with individual future single-day stock return at lags of 1, 2, 3, and 5 trading sessions using Pearson correlation.

C. Statistical validation

- Use a chronological train-test split.
- Apply false discovery rate (FDR) correction to account for multiple comparisons.
- Treat the findings as exploratory; correlations do not establish causation.

D. Portfolio construction and backtesting

- Convert sentiment signals into portfolio weights.
- Apply a maximum individual stock weight of 20%, smoothing, and a deadband.
- Account for transaction costs.
- Compare the sentiment-based strategy with an equal-weight benchmark using the same price calendar.

Sentiment score | 1 trading session | 2 trading sessions | 3 trading sessions | 5 trading sessions

RESULTS

Statistical Relationships Did Not Translate into Portfolio Outperformance

Reaction fingerprint findings

- 88 stock-signal-lag combinations evaluated in the full-sample analysis.
- Zero combinations remained statistically significant after FDR correction.
- Chronological train-test validation was used to investigate relationship stability.

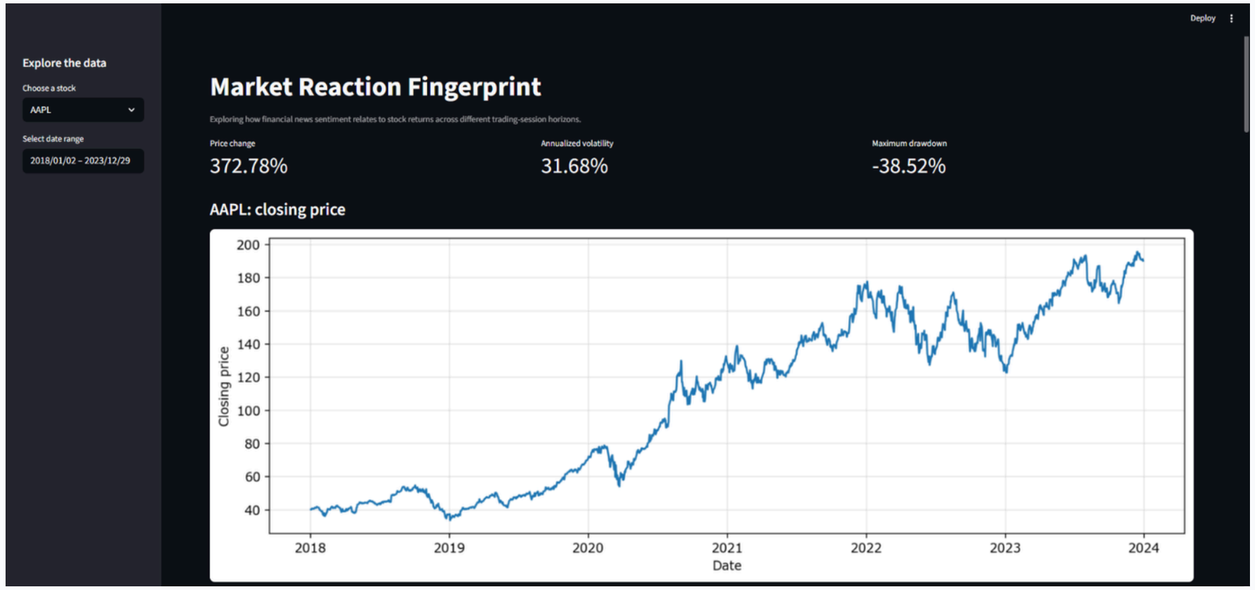
Portfolio backtest comparison

METRIC	SENTIMENT STRATEGY	EQUAL-WEIGHT BENCHMARK
Total return	-11.46%	+175.39%
Annualized volatility	17.32%	20.96%
Sharpe ratio	-0.031	0.913
Maximum drawdown	-38.15%	-33.74%
Average daily turnover	40.09%	1.16%
Average cash allocation	23.69%	0%

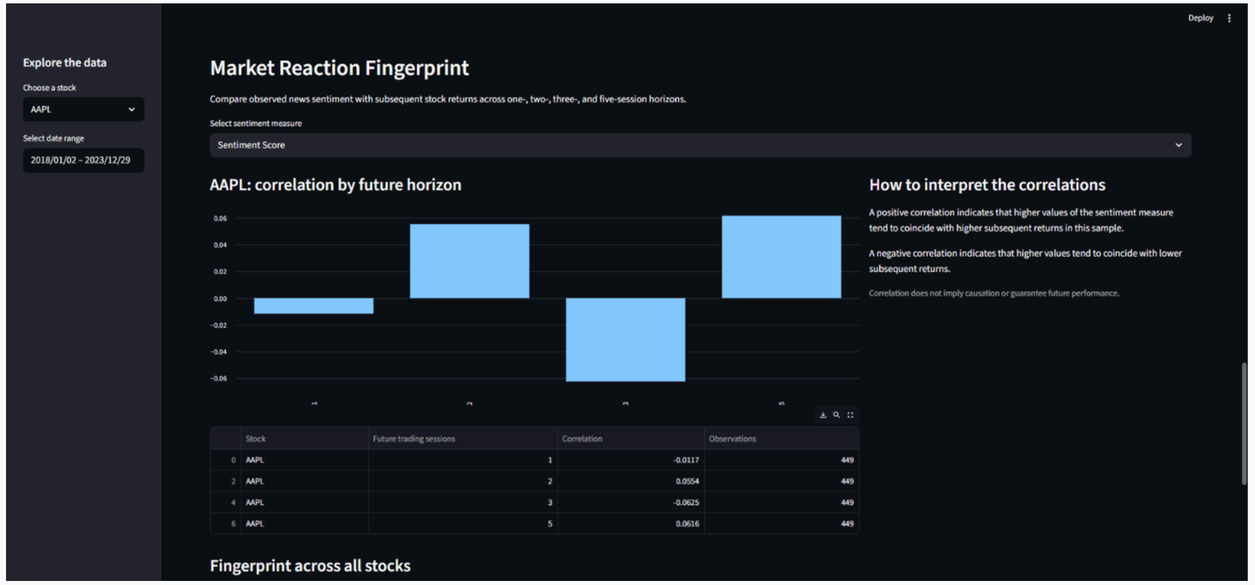
Key interpretation: The sentiment strategy underperformed the equal-weight benchmark over the evaluated period. It had lower annualized volatility but a negative Sharpe ratio and a larger maximum drawdown.
Historical backtest results; not a guarantee of future performance. Verify figures against the final results CSV before submission.

An Interactive Workflow for Financial Sentiment Analysis

Stock analytics



Reaction fingerprint



Why It Matters

- Enables comparison of sentiment-return relationships across stocks.
- Helps investigate whether observed signals are stable across time.
- Connects NLP-based analysis with portfolio-level backtesting.
- Makes limitations visible before considering practical investment use.

An analytical framework for testing sentiment hypotheses — not an assumption that news sentiment predicts profitable trades.

Test the Hypothesis, Validate the Signal

What we learned

- None of the 88 full-sample combinations remained significant after FDR correction.
- The sentiment-based portfolio underperformed the equal-weight benchmark.
- Lower volatility alone did not translate into better risk-adjusted performance.

Current limitations

- Incomplete or unreliable publication-time information limits precise news-to-market alignment.
- Correlation does not establish causation.
- Financial time-series dependence may affect conventional significance tests.
- The current prototype focuses on sentiment analysis and portfolio backtesting; it is not a complete portfolio stress-testing engine.

Next steps

- Improve timestamp alignment and evaluate methods that account for time-series dependence.
- Test the analysis on additional unseen periods and datasets.
- Explore alternative portfolio construction and risk controls.
- Develop explicit event-driven portfolio stress scenarios to strengthen alignment with Module **B**.

The goal is not to assume that news predicts the market, but to test whether stock-specific sentiment patterns provide measurable value.

Thank you